

10.3 Creating Displays from Survey Data

Lesson Introduction

 Benchmarks	MA.6.DP.1.1 Recognize and formulate a statistical question that would generate numerical data. MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data. MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range. MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.			 Learning Targets	- I can create a statistical question and collect survey data. - I can determine ways to collect truthful data. - I can create box plots and histograms to display data. - I can describe and interpret data displays representing a real-world context.
 Benchmark Coherence	Building On MA.5.DP.1.1 Collect and represent numerical data, including fractional and decimal values, using tables, line graphs, or line plots.		Working Toward MA.7.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation.		
 Essential Questions	- What makes a good statistical question to collect survey data? - What real-world context can surveys be used for? - Why is it important to collect truthful data? - What is the best way to ensure data that is collected is truthful?				
 Lesson Narrative	In this lesson, students apply what they have learned in the previous two lessons and the prior unit to create survey questions and collect data from the class. Students begin by generating three survey questions to collect quantitative data, then work in small groups to choose one to survey the class. Once data has been collected, students display the data in several ways and describe the distribution before summarizing the results using statistical terminology.				
 Component Summary	10.3.1 Warm-Up (~5-10 min.)	Create survey questions to gather quantitative data (<i>SE p. 217</i>)	10.3.3 Activity (15-20 min.)	Survey the class using questions from the collaboration and display class results using statistical terminology (<i>SE p. 218</i>)	
	10.3.2 Collaboration (10-15 min.)	Determine ways to collect truthful data with the survey questions from the Warm-Up (<i>SE p. 217</i>)	10.3.4 Wrap-Up (~5 min.)	Reflect on understanding of learning targets (<i>SE p. 219</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Quantitative data - Interquartile range - Quartile - Histogram		(optional) Computers (optional) Measuring tape		
- Consider creating a way to group students before they begin the lesson using formative assessment data from previous lessons. - If using Google Forms or Microsoft Forms, consider showing students how create a survey before the lesson.					

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse quantitative and qualitative data. - Students may use “yes” or “no” questions instead of questions that garner more variability in responses. - Students may confuse the interquartile range with the entire range. - Students may incorrectly calculate the minimum and the maximum with the first quartile and the third quartile.	- Students are to use the questions created in the 10.3.1 Warm-Up in the 10.3.2 Collaboration. - Consider grouping students by different questions in the Collaboration. - Consider checking student survey questions before moving on to the 10.3.3 Activity to ensure surveys will be successful and classroom appropriate.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 10.3.3 Activity positions students to begin to notice trends in the data they have collected from the class. Engage the small groups in a Notice and Wonder routine to encourage students to make observations and predictions about their data before formalizing their findings.	MA.K12.MTR.7.1: Mathematics in Real-World Contexts This lesson is a real-world application of mathematics students have been learning in this unit. Students connect mathematical concepts to their experiences as they perform their own survey to collect and analyze data. Consider discussing with students how the concepts they practice in this lesson can be utilized in other disciplines, such as science.
 Differentiate Instruction	This lesson is uniquely positioned to engage learners through student choice. Students are able to create their own survey questions, providing entry points for all learners and encouraging authentic investment in the content of the lesson. 10.3.3 Activity also provides opportunities for student choice, as students work together to create various displays for their data and analyze and summarize the data sets.	
Lesson Notes:		

10.3.1 Warm-Up: Creating Statistical Questions

Component Overview: Students will create survey questions that will gather quantitative data points. These questions will be used in the Collaboration. Consider giving students time to revise their questions after further scaffolding, as they play an important role throughout the lesson.



10.3.1 Warm-Up: Creating Statistical Questions

Create three statistical questions that could be used in a class survey to collect quantitative data.

Answers will vary.

1. *How many siblings do you have?*
2. *How far do you live from school?*
3. *How long do you spend getting ready in the morning before school?*



Consider asking students to list their interests and using these to create statistical questions. Utilizing contexts students are familiar with provides additional entry points and means of engagement in the lesson.



Students may write questions that yield categorical data. Ask those students how they would display the data in a box plot to help them identify their mistake.

10.3.2 Collaboration: Determining Ways to Collect Truthful Data

Component Overview: Students will work with a group to choose the best three statistical questions from their Warm-Up. Students then consider ways in which they can collect truthful data to answer each question and explain why the questions they chose are good statistical questions.



10.3.2 Collaboration: Determining Ways to Collect Truthful Data

Form a group of 3-4 classmates and discuss the questions everyone in your group created in the Warm-Up.

1. Choose the best three statistical questions that will produce quantitative data.

Statistical Question	Ways to Collect Truthful Data
1. <i>Answers will vary. How long do you spend getting ready in the morning before school?</i>	<i>Have each student time themselves the following morning.</i>
2. <i>How far do you live from school?</i>	<i>Record the distance from each student's home to school using Google Maps.</i>
3. <i>What is your height?</i>	<i>Measure each student's height with a measuring tape.</i>

2. With your group, discuss ways to collect truthful data for each of the questions. Summarize your discussion in the table for each statistical question.
Shown in table.
3. Explain what makes these good statistical questions.
Answers will vary.
Students will have different responses to the questions. The responses can be based on measurement and data, instead of self-reporting.



Consider using formative data gathered in previous lessons to create groups of students ahead of time. Group students who may struggle with students who have demonstrated understanding.

10.3.3 Activity: Class Survey

Component Overview: In this activity, groups choose one question to use to survey the rest of the class. Display the questions around the room for the class to access. Groups can rotate to each survey question and record their name and their response, then move on to the next question when given the signal. Alternatively, groups can take turns surveying the class through a show of hands and recording the responses on their data chart. Students then display the data set they collected in a table, create a data display to accompany the data set, and describe measures of variation and center to summarize the results.



10.3.3 Activity: Class Survey

Data Collection Directions:

1. As a group, choose one question to survey the class.
2. Write your group's question in the top of the data chart.
3. Place the data chart in a location easily accessible by your classmates. Your teacher may have further instructions.
4. Follow your teacher's directions to answer each group's question.

Answers will vary.

My Group's Statistical Question:			
Name	Answer	Name	Answer
1.		15.	
2.		16.	
3.		17.	
4.		18.	
5.		19.	
6.		20.	
7.		21.	
8.		22.	
9.		23.	
10.		24.	
11.		25.	
12.		26.	
13.		27.	
14.		28.	



Allowing students to create their survey in a digital format such as Google Forms or Microsoft Forms will provide a more streamlined way to collect the data. A digital format also provides additional entry points for students who may struggle to organize their work or with writing.



Notice and Wonder

Once the data has been collected by each group, engage the groups in a Notice and Wonder routine to position students to begin to notice trends in the data set as well as consider any questions they have after considering the data. Circulate as students discuss and encourage groups to connect their notices to the statistical terminology they will be using in the rest of the component.

10.3.3 Activity: Class Survey (continued)

Once the results are collected from the whole class, work with your group to create a presentation displaying the data from your survey results.

Data Display Directions:

- Present the data in a neat, organized table.
- Find the five-number summary and create a box plot.
- Create a frequency table and a histogram.
- Describe the distributions shape, center, spread, and unusual features.
- Summarize the class' results using statistical terminology.

Answers will vary.



It may be helpful to assign each point of the data display to different students or have students choose to ensure each student participates equally. This will provide entry points for all students and encourage participation and discussion throughout the component. Consider splitting the first three points among the group but encouraging all students to work on the last two bullet points together while also verifying that the data displays are accurate in their group.



Students may struggle to correctly create each data display and use them to synthesize the information collected during the survey. Ask questions to further scaffold this component:

- *What is the five-number summary? What information must be included?*
- *How can you use the data table to create data displays?*
- *What is IQR and how do you find it? Which data display shows IQR?*



If groups finish early, challenge them to make predictions about the other data sets that were collected. Will the other groups' display look like theirs? Why or why not?

10.3.4 Wrap-Up: Reflection

Component Overview: Students will reflect on their understanding of different parts of the lesson. Students will use the given icons to communicate their understanding. Encourage students to reflect thoughtfully to ensure any looming questions are answered.



10.3.4 Wrap-Up: Reflection

- Circle the icon that best describes your understanding of each topic.

Answers will vary.



I need help.



I got it.



I could help someone.

Determining ways to collect truthful data			
Finding a five-number summary			
Creating a box plot			
Creating a frequency table			
Creating a histogram			
Describing data displayed on a box plot, frequency table, and histogram			



Consider allowing students to write down any questions they may have about the options they circled the “I need help” icon for on a sticky note and placing it in a designated area for the teacher to answer the following day.